

Krishnaa Vadivel | Computational Physics PhD | First-Principles Materials | HPC | Scientific ML

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Profile

Computational physics PhD researcher with training across electrical engineering, condensed matter, semiconductor physics, numerical methods, and scientific software. Current work centers on first-principles calculations of self-trapped excitons and excitonic polarons, many-body formulations of electron-phonon coupled systems, and scalable implementations for national-lab HPC platforms.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Develop first-principles research workflows for self-trapped excitons, excitonic polarons, and related excited-state phenomena in materials, linking DFT, DFPT, GW/BSE-style ideas, and effective Hamiltonian viewpoints.
- Study many-body and field-theoretic formulations of coupled electron, exciton, and phonon systems, including coherent-state and path-integral approaches for time-dependent and self-localized states.
- Implement distributed scientific computing pipelines in Python and C++ with MPI, OpenMP, PETSc, and SLEPc for large linear-algebra and simulation workloads.
- Port and optimize computational workloads for accelerator-enabled HPC environments using CUDA and ROCm, with experience running on Frontier, Perlmutter, and other leadership-class clusters.
- Investigate machine-learning accelerators for scientific computing, including equivariant graph neural networks for molecular-property prediction and ML-assisted materials workflows.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-based data-processing pipelines and custom OpenGL visualization software for large acoustic waveform datasets generated by downhole logging tools.
- Developed embedded software and digital logic for sensor logging and tool control using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-based systems.
- Supported hardware-oriented engineering tasks including circuit-level debugging, oscilloscope-based validation, field testing, and targeted PCB modifications in Altium.
- Contributed host-side engineering software with C#, ASP-based services, and Microsoft SQL-backed database workflows.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical articles and product-facing photonics content covering lasers, optics, and optoelectronic components for a specialized engineering readership.

Selected Projects

Distributed First-Principles Exciton-Phonon Simulation

- Developed computational workflows for first-principles excited-state and exciton-phonon calculations, with an emphasis on self-trapped excitons, scalable many-body formulations, and reusable HPC pipelines.

Semiconductor Device Simulation

- Built numerical treatments of semiconductor transport and electrostatics, including drift-diffusion and pn-junction-style device equations, to connect condensed-matter models with engineering-scale simulation.

Equivariant Graph Neural Networks for Molecular Systems

- Implemented graph-based and symmetry-aware ML models for molecular-property prediction and explored their use as accelerators for scientific workflows.

Agentic Scientific Input Generation

- Built retrieval-driven tooling that reads software documentation and research papers to generate structured computational-science inputs and starter workflows for physics and chemistry codes.

Integrated Photonics and FDTD Modeling

- Simulated photonic structures including band-pass and resonator-style devices with FDTD tooling such as Meep.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing doctoral research in computational materials, many-body theory, and scientific computing.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included work in stochastic computing, device-oriented EE topics, and advanced design coursework.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included embedded systems, robotics, and core mathematics, physics, and computing foundations.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society*, 147(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib, mpi4py, petsc4py, slepc4py

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Systems: Linux command line, macOS, Windows, CMake, Docker, Docker Compose, Kubernetes, gcloud

Hardware: Lattice FPGA, PIC32, dsPIC, STM32, C8051, PCB bring-up and instrumentation-based debugging

HPC Platforms: Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments